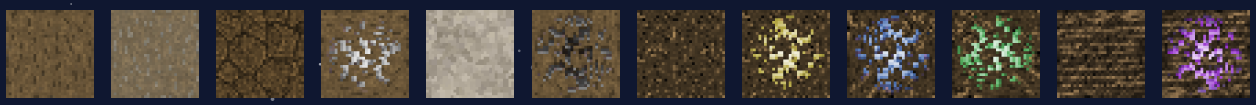
SPACEBOX · DESIGN PROPOSAL

New Crafts & Blocks

Ideas for recipes, processing chains, blocks and items that fit the current ore tiers and progression



The twelve natural blocks of the Default GameSet, from drill power 1 to 5.

Based on game version 0.1.5, Default GameSet

85 blocks · 113 items · 159 crafting blueprints · 63 processing recipes

October 2026 · Discussion document, nothing in it is implemented

SECTION 1

Summary

Spacebox already has a complete early and mid game: the aluminium drill and furnace lead to cobalt energy cells, iron machines, thulium optics, titanium and uranium tools, nuclear power, ships with cannons and an obscurium end tier. This document looks at where that content runs out and proposes **37 ideas** in seven groups. Each one uses existing items and ores where it can, names the behaviour it needs (data-only JSON versus a new `Block` subclass) and gets an effort and priority estimate.

37	9	14	13
ideas in 7 groups	priority-1 ideas	data-only or nearly (JSON + textures)	content gaps found

Key findings

- **The last drill unlocks nothing.** The hardest natural blocks need drill power 5, so the tier-6 obscurium drill is only a speed upgrade. A tier-6 ore (G1) gives the end game its own material.
- **Automation stops halfway.** Pipes, storages and machines exist, but mining, crafting, sorting and disposal are manual. The drill rig (D1), sorter pipes (D3) and assembler (D2) close the loop.
- **Power is free and cannot be stored.** There is no battery and no generator has a running cost. A battery bank (C1) and a hydrogen fuel cell (C2) turn energy into something to plan.
- **Several systems are built but unused:** weapon ammo, the furnace fuel slot, the slab shape, the iron lens and the titanium crate (no blueprint). These are cheap wins.
- **Processing chains are one step long** because every machine recipe has a single input and output. An optional byproduct field (B1) and multi-input recipes (B3) open up longer chains.

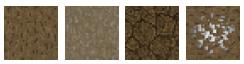
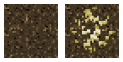
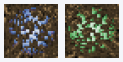
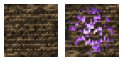
Legend

Effort	Meaning	Priority	Meaning
XS	JSON and textures only, under a day	P1	Fixes a clear gap or unlocks a core pillar; do next
S	Small subclass or a new optional JSON field, 1 to 2 days	P2	Strong value, after P1 items or alongside them
M	New block behaviour with UI, save data and sync, 3 to 5 days	P3	Polish or long-term; nice to have
L	New system across several files, 1 to 3 weeks		

SECTION 2

Current progression at a glance

Drill tier, what it can mine and what the player builds with it

Tier	Drill	New blocks it mines	What it unlocks today	Proposed additions
T1	`aluminium_drill`		Start kit: drill, furnace, crafting table, flashlight. Aluminium ingots and panels, cobalt energy cells, microchips, glass, water, oxygen and power items.	**A3 charge, A4 bricks, C3 solar**
T2	`cobalt_drill`		Iron: basic machine, crusher, iron armor, pipes, cannons, iron panels. Dense ice gives methane, then polymers and plastic.	**B1 electrolyzer, B2 reactor, C1 battery, D3 sorter**
T3	`iron_drill`		Thulium (crusher only): lenses, thulium energy cells, analyzer, radar, enhanced furnace and crusher, triple iron drill.	**C2 fuel cell, D1 drill rig, B6 sifter, F1 gyroscope**
T4	`triple_iron_drill`		Titanium armor and panels, uranium rods, nuclear power, titanium drill, ion thrusters, heavy cannons, bombs.	**A1/A2 titanium logistics, B4 centrifuge, F4 mining laser, E1/E3**
T5	`titanium_drill`		Obscurium: obscurium drill, safe, gravity generator, impulse rifle, anti-frigate cannon.	**B3 alloys, F5 composite hull, G2 obscurium tier**
T6	`obscurium_drill`	—	**Nothing new to mine.**	**G1 xenite crystal, G3 jump drive**

Drill tier compared to the block's `PowerToDrill` in blocks.json. Ore textures are taken from the Default GameSet. Thulium ore cannot be smelted directly, which teaches the crusher and is worth keeping.

Where each material goes today

Material	Main uses today	Observation
Aluminium	panels, cables, energy cells, microchips, armor, flashlights, thrusters	Well used
Cobalt	energy cells, microchips, lenses, bombs, paintings	Well used
Iron	basic machine, panels, pipes, armor, guns, cannons	Well used; iron lens has no use
Thulium	lenses, thulium energy cells, high-tier machines	Good bridge material
Titanium	panels, armor, drill, ion thruster, cannons	Titanium crate cannot be crafted
Uranium	rods for nuclear blocks, drills, bombs, cannons	Rods are a one-time cost; dust only feeds a torch
Obscurium	safe, obscurium drill, impulse rifle, gravity generator, anti-frigate cannon	Few sinks for the rarest ore
Rock, sand	sand, glass	Bulk material with almost no value
Ice, methane	water, oxygen, hydrogen, polymers	All hand-crafted; no automation path

SECTION 3

Gaps in the current content

What the JSON data and the block code show, and which ideas address each gap

#	Gap	Evidence	Addressed by
1	**Tier-6 drill has no tier-6 target**	Ores and rocks need drill power 1 to 5. The obscurium drill (tier 6) unlocks no new block, so the end of the drill line is not a goal.	**G1, G2**
2	**Uncraftable and dead-end content**	basic_titanium_crate has no blueprint and uses aluminium textures. iron_lens is only crafted and crushed back. paint has no recipe. beer and laser_gun come only from loot or creative.	**A1, C3, F4, E4**
3	**No storage progression**	All three crates are 8x3. The safe is 3x3 and costs obscurium. Better metals bring nothing for storage.	**A1, D-group**
4	**Rock and sand have little value**	Rocks are the bulk of every asteroid but only become sand, and sand only becomes glass. Heavy rock needs a tier-5 drill for the same sand.	**A4, B6, D4**
5	**Single-ingredient processing**	Every furnace and crusher recipe has one input and one output. Alloys, byproducts and multi-output chains are not possible, which keeps chains short.	**B1, B3, B6**
6	**Life support is hand-crafted**	Oxygen, liquid hydrogen and power items come from the crafting table. Machines are optional and pipes cannot feed a self-running base.	**B1, C2, D2**
7	**Power is free and cannot be stored**	Generators have no running cost after crafting and there is no battery. Bursty loads (thrusters, cannons, gravity) cannot be buffered, and energy never needs planning.	**C1, C2, C4**
8	**Automation stops at pipes**	Pipes, storages and machines exist, but there is no automatic mining, crafting, sorting by destination or disposal. DONE.txt names automation as a core goal.	**D1 to D4**
9	**Exploration rewards are thin**	Chest loot tables give one med kit. The structure definitions are empty. Parasites drop rocks.	**A8, A9**
10	**Equipment stops at titanium**	No fourth tier, no oxygen-capacity upgrade and no movement upgrade, so obscurium and uranium do little for the player personally.	**E1, E3, G2**
11	**Ships have no mining role and miss WP10 parts**	Gyroscope, landing gear and docking connectors are open WP10 items. Ships cannot mine, so they do not feed the economy.	**F1 to F4**
12	**Unused systems already in the code**	WeaponItem.Ammo works but no weapon uses it. ResourceProcessingBlock has a saved FuelStorage that nothing reads. The built-in slab shape has no survival block.	**A5, A10, B5**
13	**Ambiguous recipes**	Red and orange torches use the same dust, and so do cyan and pink.	**A7**

What works well and should be kept

- The drill line requires the previous drill in every recipe, so each tier builds on the last.
- Furnace without power, crusher with power: the first machine is free and the second one teaches the power grid.
- Crusher recycling never returns more metal than the craft costs (checked by tests), so no ideas below create an exploit loop.
- Descriptions carry no numbers and parameters sit in tooltips; every new block or item below should follow that rule.

SECTION 4

Ideas overview

All ideas at a glance, grouped by area

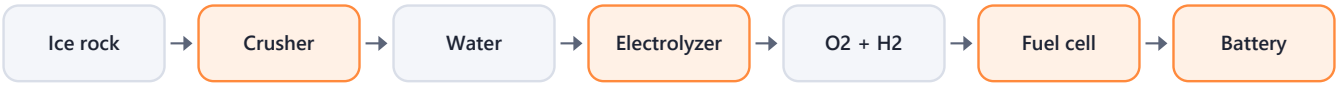
ID	Idea	Kind of change	Effort	Prio
A1	Titanium Crate blueprint and storage tiers	JSON: blueprint + storage blocks	XS	P1
A2	Fast Titanium Pipes	JSON: pipe blocks	XS	P1
A3	Mining Charge	JSON: explosive block	XS	P1
A4	Regolith Bricks	JSON: building blocks	XS	P2
A5	Metal Slabs and Catwalk Grating	JSON: slab model blocks	S	P2
A6	Tinted and Armored Windows	JSON: window blocks	XS	P3
A7	Distinct Light Dyes	JSON: recipe change	XS	P3
A8	Nano Med Kit and Parasite Gland	JSON: items, loot, recipes	XS	P2
A9	Loot Table Overhaul	JSON: loot tables	S	P2
A10	Ammunition for Kinetic Weapons	JSON: weapon + ammo items	S	P3
B1	Electrolyzer	Processing subclass + Byproduct field	S	P1
B2	Chemical Reactor	Processing subclass	S	P2
B3	Alloy Smelter	Multi-input processing (code)	M	P2
B4	Enrichment Centrifuge	Processing subclass + items	S	P2
B5	Furnace Fuel Slot	Furnace code (fuel slot)	M	P3
B6	Regolith Sifter	Processing subclass + chance byproducts	S	P2
C1	Battery Bank	New electrical block type	M	P1
C2	Fuel-burning Generators	Generator Fuel section (code)	M	P1
C3	Solar Panel	JSON generator, later code	XS	P2
C4	Power Switch and Power Monitor	Two small block types	M	P2
D1	Automated Drill Rig	New multi-cell block type	L	P1
D2	Assembler	New container block type	M	P2
D3	Sorter and Overflow Pipes	PipeBlock mode option	S	P2
D4	Incinerator	Small endpoint block	S	P3
E1	Oxygen Tank Upgrades	Equipment stat field	S	P2
E2	Medical Bay	New consumer block	M	P3
E3	Jetpack Booster	New equipment slot + movement	M	P2
E4	Hydroponics Bay	Processing subclass + light check	M	P3
F1	Gyroscope	New ship block type	M	P1
F2	Magnetic Landing Gear	Ship physics joint	M	P2
F3	Docking Connector	Cross-entity joint + networks	L	P3
F4	Mining Laser Cannon	Projectile mining flag	M	P2
F5	Composite Hull Blocks	JSON: hull blocks (after B3)	XS	P2
G1	Tier-6 Ore: Xenite Crystal	JSON: ore, items, generation	S	P1
G2	Obscurium Equipment Tier	JSON: equipment items	S	P2
G3	Jump Drive	New ship system	L	P3
G4	Ore Scanner Mode	Radar fields + background scan	M	P2

SECTION 5

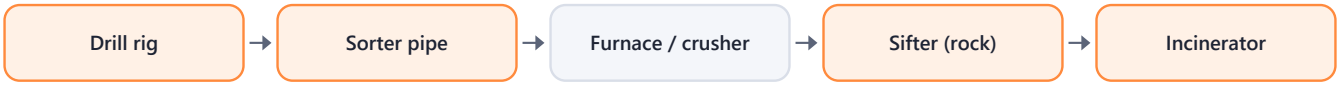
New production chains

How the ideas connect into longer chains (highlighted = new)

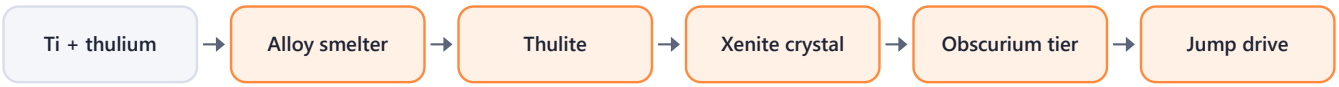
Life support loop (B1, C1, C2)



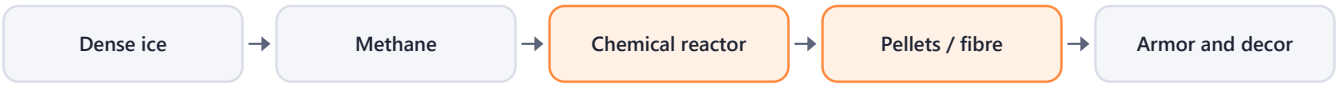
Automated mining line (D1, D3, B6, D4)



Alloy and end-game chain (B3, F5, G1, G2, G3)



Polymer line (B2)



Byproducts without breaking saves

Several ideas (B1 electrolyzer, B6 sifter) need a second output. The smallest change is an optional field on a processing recipe. A recipe without it behaves as it does now, so recipes.json and old machine saves stay valid:

```
{ "DurationSec": 8.0, "PowerPerTickRequared": 10,
  "Ingredient": { "Item": "water", "Quantity": 1 },
  "Product":    { "Item": "compressed_oxygen", "Quantity": 1 },
  "Byproduct": { "Item": "liquid_hydrogen", "Quantity": 1, "Chance": 100 } }
```

The machine needs a second output slot (a 1x2 OutputStorage), the pipe endpoint keeps taking from any output slot, and the disassembler ignores byproducts when it derives recipes. New machine types (electrolyzer, reactor, centrifuge, sifter) are thin `ResourceProcessingBlock` subclasses registered in `BlockFactory`, because recipes are looked up by the block's `Type`.

SECTION 6.1

A. Quick wins: data-only content

Pure GameSet JSON plus textures. No code, no save-format change, shippable in a patch.

A1 Titanium Crate blueprint and storage tiers

PRIORITY P1 EFFORT XS

IDS	`basic_titanium_crate (existing), large_iron_crate, large_titanium_crate`
PURPOSE	basic_titanium_crate exists in blocks.json and disassembly.json, but no blueprint makes it and it uses the aluminium crate textures. All three crates hold 8x3, so better metal buys nothing.
PROGRESSION	Gives iron and titanium a storage payoff. Suggested: titanium crate 8x3 with high durability for ships, and new large crates (8x4 iron, 8x6 titanium) for bases.

RECIPE SKETCH	`basic_titanium_crate <= titanium_panels x3`
	`large_iron_crate <= iron_panels x6, basic_iron_crate x1`
	`large_titanium_crate <= titanium_panels x6, basic_titanium_crate x1, microchip x1`

BEHAVIOUR	Data-only (Type storage, StorageSize). Needs titanium crate textures. New ids rather than resizing existing crates, so old saves keep their slot layout.

A2 Fast Titanium Pipes

PRIORITY P1 EFFORT XS

IDS	`titanium_item_pipe, titanium_item_pipe_straight, titanium_item_pipe_merger`
PURPOSE	Pipes move one item per second. A furnace chain fed by a drill rig or several crushers saturates that quickly.
PROGRESSION	Late-game logistics tier after the iron pipes from the iron age chapter.
RECIPE SKETCH	`titanium_item_pipe x2 <= titanium_panels x2, microchip x1, thulium_energy_cell x1` `JSON: Pipe { TransferInterval 0.25, Power 12 }`
BEHAVIOUR	Data-only. PipeBlock already reads TransferInterval, Power, InputSides and OutputSides.

A3 Mining Charge

PRIORITY P1 EFFORT XS

IDS	`mining_charge`
PURPOSE	Both bombs use DropBlocks false, so explosives are weapons only. A small charge that drops the blocks it breaks turns explosives into a mining tool.
PROGRESSION	Mid game (iron age). Gives compressed methane and cobalt dust a use outside polymers and paintings.
RECIPE SKETCH	`mining_charge x2 <= cobalt_dust x2, compressed_methane x1, iron_panels x1` `JSON: Explosive { Radius 2, Penetration 25, Damage 10, DropBlocks true, PushStrength 4 }`
BEHAVIOUR	Data-only (Type explosive). Check the ExplosionBlockBreaker merged-drop path with DropBlocks true and the claim and gamerule settings.

A4 Regolith Bricks

PRIORITY P2 EFFORT XS

IDS	`regolith_brick, regolith_brick_slab`
PURPOSE	Light, medium and heavy rock only become sand, and sand only becomes glass. Players end up with chests full of rock.
PROGRESSION	Cheap tier-0 building material for early bases, before aluminium panels are worth spending on walls.
RECIPE SKETCH	`regolith_brick x4 <= sand x4, water x1` `regolith_brick_slab x2 <= regolith_brick x1 (Model slab)`
BEHAVIOUR	Data-only. Mass 6, Durability 25, PowerToDrill 2. Footsteps can use the stone set.

A5 Metal Slabs and Catwalk Grating

PRIORITY P2 EFFORT S

IDS	`aluminium_slab, iron_slab, titanium_slab, iron_grating`
PURPOSE	The engine has a built-in slab shape (BlockGeometryLibrary) that no survival block uses. Gravity rooms and ship interiors need half blocks and see-through floors.
PROGRESSION	Building variety in every metal tier; grating is useful now that crouching and gravity rooms exist.
RECIPE SKETCH	`aluminium_slab x2 <= aluminium_panels x1 (same pattern for iron and titanium)` `iron_grating x4 <= iron_panels x2`
BEHAVIOUR	Data-only (Model slab; grating IsTransparent with a cut-out texture). Crusher recycling must stay at or below the panel cost (DefaultCrusherRecipesTests).

A6 Tinted and Armored Windows

PRIORITY P3 EFFORT XS

IDS	`thulium_window, armored_window`
PURPOSE	Glass has two 1x1 tiers and the big window. Ships need a window that survives cannon fire.
PROGRESSION	Thulium (tier 3) gives a coloured window; titanium (tier 4) gives an armored one.
RECIPE SKETCH	`thulium_window x2 <= glass x3, thulium_dust x1` `armored_window x1 <= reinforced_window x1, titanium_panels x2`
BEHAVIOUR	Data-only (IsTransparent, Durability 40 for armored, glass footsteps).

A7 Distinct Light Dyes

PRIORITY P3 EFFORT XS

IDS	`light_orange, light_pink (recipe change only)`
PURPOSE	Red and orange torches are both light_white + iron_dust; cyan and pink are both light_white + cobalt_dust. Players cannot guess which dust makes which colour.
PROGRESSION	Gives obscurium and aluminium dust a decorative use.
RECIPE SKETCH	`light_orange <= light_white x1, iron_dust x1, aluminium_dust x1` `light_pink <= light_white x1, obscurium_dust x1`
BEHAVIOUR	Data-only recipe change. The tutorial only places light_white and rewards light_blue, so it is not affected.

A8 Nano Med Kit and Parasite Gland

PRIORITY P2 EFFORT XS

IDS	`parasite_gland, nano_med_kit`
PURPOSE	The med kit is the only healing item. Parasites drop rocks that a drill gives faster, so fighting them has no reward.
PROGRESSION	The parasite loot table gets a 25 % gland drop; the gland is used in a stronger med kit and an alternative DNA sample recipe.
RECIPE SKETCH	`nano_med_kit x1 <= med_kit x2, parasite_gland x1, polymer_fiber x1 (HealAmount 100)` `dna_sample x1 <= parasite_gland x2, water x1, glass x1 (no $health cost)`
BEHAVIOUR	Data-only (consumable item, Loot/mineral_parasite.json, crafting.json).

A9 Loot Table Overhaul

PRIORITY P2 EFFORT S

IDS	`Loot/chest.json, Loot/alien-chest.json, spacer.json`
PURPOSE	chest and alien-chest both drop one med kit only, and the structure files are empty. Exploring has no payoff.
PROGRESSION	Chests hold early-game boosts (microchips, energy cells, iron drill bits); alien chests hold rare tier-3 to tier-4 items (thulium lenses, uranium rods, obscurium dust, A8 glands).
RECIPE SKETCH	`No crafting. Example alien-chest: microchip 60 % x1-3, thulium_lens 40 % x2-4, uranium_rods 15 % x1-2, obscurium_dust 10 % x1-2`
BEHAVIOUR	Data-only. A follow-up is real content for abandoned_ship, alien_base and ruins.

A10 Ammunition for Kinetic Weapons

PRIORITY P3 EFFORT S

IDS	`iron_slug, mass_driver`
PURPOSE	WeaponItem.Ammo is implemented in InteractionShoot but every weapon uses an empty Ammo field. All guns run on suit power only.
PROGRESSION	A new heavy kinetic gun with real ammo gives iron a steady use and a reason to automate a slug line. Existing guns stay power-only, so nothing breaks.
RECIPE SKETCH	`iron_slug x8 <= iron_ingot x1` `mass_driver x1 <= titanium_ingot x6, microchip x2, thulium_energy_cell x4 (Ammo iron_slug, PowerUsage 2)`
BEHAVIOUR	Data-only (weapon item, projectile entry, ammo item).

SECTION 6.2

B. Processing chains and new machines

New processing machine types. Most reuse ResourceProcessingBlock: a tiny subclass, one BlockFactory entry and a new RecipeCatalog type.

B1 Electrolyzer

PRIORITY P1 EFFORT S

IDS	`electrolyzer (Type electrolyzer)`
PURPOSE	Water becomes oxygen and hydrogen only by hand at the crafting table (compressed_oxygen, liquid_hydrogen) or in the furnace. A base cannot make oxygen and fuel on its own.
PROGRESSION	Iron age. Pipe-fed with water from a crusher and ice line; feeds the oxygen generator, the fuel cell (C2) and thruster crafting.
RECIPE SKETCH	`electrolyzer <= basic_machine x1, iron_panels x4, cobalt_lens x2, energy_cell x2` `water x1 -> compressed_oxygen x1 + byproduct liquid_hydrogen x1 (8 s, 10 power per tick)`
BEHAVIOUR	Small subclass of ResourceProcessingBlock plus an optional Byproduct field on processing recipes (default none, so recipes.json stays compatible). Keep the hand recipes, because tutorial chapter 5 crafts compressed oxygen and power by hand.

B2 Chemical Reactor

PRIORITY P2 EFFORT S

IDS	`chemical_reactor (Type reactor)`
PURPOSE	Polymers come from hand crafting (methane + 2 water -> 6 pellets) and the furnace (4 pellets -> plastic, 20 s). Armor, paintings and frames need a lot of fibre.
PROGRESSION	Mid game. Automates the polymer line and gives methane ice a powered, better-yield route.

RECIPE SKETCH	`chemical_reactor <= basic_machine x1, iron_panels x4, thulium_energy_cell x2, glass x4`
	`compressed_methane x1 -> polymer_pellets x8 (12 s)`
	`polymer_pellets x4 -> polymer_fiber x2 (10 s)`
	`polymer_pellets x4 -> plastic x1 (10 s)`

BEHAVIOUR	Small subclass of ResourceProcessingBlock, new RecipeCatalog type reactor.

B3 Alloy Smelter

PRIORITY P2 EFFORT M

IDS	`alloy_smelter, cobalt_steel_ingot, thulite_ingot, obscurium_alloy_ingot`
PURPOSE	Every processing recipe has exactly one ingredient, so the ores never combine. The jump from titanium to obscurium is steep: obscurium drill, safe, gravity generator and anti-frigate cannon all use raw obscurium ingots.
PROGRESSION	Adds intermediate alloys: cobalt steel (tier 3, drills and pipes), thulite = titanium + thulium (tier 4, armor, thrusters) and obscurium alloy (tier 5, end-game hull and tools).

RECIPE SKETCH	`alloy_smelter <= enhanced_furnace x1, titanium_panels x6, microchip x4`
	`iron_ingot x2 + cobalt_ingot x1 -> cobalt_steel_ingot x2`
	`titanium_ingot x2 + thulium_ingot x1 -> thulite_ingot x2`
	`obscurium_ingot x1 + titanium_ingot x2 -> obscurium_alloy_ingot x2`

BEHAVIOUR	Needs multi-ingredient processing recipes (Ingredients list with the old single Ingredient still read), a second input slot in ResourceProcessingGUI, pipe insertion rules and serializer support. Medium code change.

B4 Enrichment Centrifuge

PRIORITY P2 EFFORT S

IDS	`centrifuge, enriched_uranium, enriched_uranium_rods`
PURPOSE	Uranium dust only becomes ingots or a green torch. Nuclear power has a single fuel quality.
PROGRESSION	Tier 4 to 5. Enriched rods feed C2 generator fuel and a new top-tier reactor, and replace plain rods in the obscurium drill.

RECIPE SKETCH	`centrifuge <= basic_machine x1, titanium_panels x4, microchip x2, thulium_energy_cell x4`
	`uranium_dust x4 -> enriched_uranium x1 (30 s, high power)`
	`enriched_uranium_rods x2 <= enriched_uranium x1, titanium_panels x1`

BEHAVIOUR	Small subclass of ResourceProcessingBlock plus items. Keep enriched products out of the crusher, or make them return at most the dust cost (crusher balance test).

B5 Furnace Fuel Slot

PRIORITY P3 EFFORT M

IDS	`furnace, enhanced_furnace (behaviour)`
PURPOSE	ResourceProcessingBlock already has a FuelStorage that ResourceProcessingSerializer saves, but nothing uses it. The furnace needs no power, so it never asks for anything.
PROGRESSION	Optional speed-up: compressed methane or liquid hydrogen in the fuel slot gives 1.5x speed for N recipes. Without fuel the furnace works as it does now, so the tutorial does not change.
RECIPE SKETCH	`No new crafting. JSON: Furnace { Fuels [ { Item compressed_methane, Recipes 4, SpeedMultiplier 1.5 } ] }`
BEHAVIOUR	Code: read the fuel slot in the tick task, show it in ResourceProcessingGUI, let pipes insert fuel from the side. The save format already has the slot.

B6 Regolith Sifter

PRIORITY P2 EFFORT S

IDS	`sifter (Type sifter)`
PURPOSE	Heavy rock needs a tier-5 drill and only gives sand. Rock is the most common block and has almost no value.
PROGRESSION	Mid game. Sifting rock gives a small chance of trace metals, so stone mining by a drill rig (D1) still pays.

RECIPE SKETCH	`sifter <= basic_machine x1, iron_panels x4, iron_drill_bit x2`
	`medium_rock x1 -> sand x3 + 20 % iron_dust x1`
	`heavy_rock x1 -> sand x4 + 15 % titanium_dust x1 + 5 % thulium_dust x1`

BEHAVIOUR	Processing subclass plus chance byproducts (same optional Byproduct field as B1, with Chance). Balance against the crusher recycling rule.

SECTION 6.3

C. Power and electricity

Energy storage, fuel-burning generators and network control, so power becomes a resource to manage instead of a one-off craft.

C1 Battery Bank

PRIORITY P1 EFFORT M

IDS	`cobalt_battery, thulium_capacitor (Type battery)`
PURPOSE	Generators produce a constant output and nothing stores it. Thrusters, cannons and gravity generators need short bursts. TODO.txt already lists a battery texture.
PROGRESSION	Cobalt battery after the first power grid (chapter 6), thulium capacitor (1x2x1) for ships and gravity rooms.
RECIPE SKETCH	`cobalt_battery <= aluminium_panels x2, energy_cell x6, microchip x1 (capacity 5 000 EU, 40 EU/s in and out)` `thulium_capacitor <= iron_panels x4, thulium_energy_cell x8, microchip x2 (capacity 20 000 EU, 150 EU/s)`
BEHAVIOUR	New ElectricalBlock subclass and ElectricalManager support for storage nodes (charge on surplus, discharge on deficit), NBT battery_data with a default for missing data, tooltip charge bar, network sync of the charge level.

C2 Fuel-burning Generators

PRIORITY P1 EFFORT M

IDS	`hydrogen_fuel_cell, generator Fuel section`
PURPOSE	All generators produce power for free after crafting. Liquid hydrogen only feeds hand crafting, and uranium rods are a one-time cost.
PROGRESSION	A hydrogen fuel cell (90 EU) is a mid-game generator that uses the electrolyzer output. An optional Fuel section lets mods make the nuclear blocks consume rods too.
RECIPE SKETCH	`hydrogen_fuel_cell <= basic_machine x1, iron_panels x4, thulium_lens x2, energy_cell x4` `JSON: Generator { Power 90, Fuel { Item liquid_hydrogen, Interval 60, TankCapacity 20 } }`
BEHAVIOUR	Generator code reads an optional Fuel section (no section = current behaviour). Tank, UI and serializer like the oxygen generator; pipes insert into the filtered slot.

C3 Solar Panel

PRIORITY P2 EFFORT XS

IDS	`solar_panel`
PURPOSE	There is no passive low-tier generator besides the cobalt block. iron_lens has no use besides recycling.
PROGRESSION	Early to mid game. Gives the iron lens a purpose and a silent, safe source for remote outposts and claim beacons.
RECIPE SKETCH	`solar_panel <= glass x4, aluminium_panels x2, iron_lens x2, energy_cell x1`
BEHAVIOUR	Stage 1: data-only generator (Power 10). Stage 2: output scales with sun exposure (raycast toward the skybox sun, refreshed every few seconds).

C4 Power Switch and Power Monitor

PRIORITY P2 EFFORT M

IDS	`power_switch, power_monitor`
PURPOSE	A network cannot be split without breaking cables, and the player cannot see production versus consumption without opening every block.
PROGRESSION	Switch: turn machines or gravity off; a ship console can bind it like other trigger blocks. Monitor: a sign-like screen that shows network output, load and battery charge.
RECIPE SKETCH	`power_switch x2 <= iron_panels x1, cable_block x2, microchip x1` `power_monitor <= iron_panels x2, glass x1, microchip x2`
BEHAVIOUR	Switch: CableBlock variant that passes or blocks the network. Monitor: SignBlock-style renderer fed from ElectricalManager stats. Both need network sync.

SECTION 6.4

D. Logistics and automation

Closing the automation loop that DONE.txt lists as a goal: mine, move, sort, craft and dispose without the player.

D1 Automated Drill Rig

PRIORITY P1 EFFORT L

IDS	`drill_rig (Type drillrig, 2x2x2)`
PURPOSE	Automation is the first goal in DONE.txt, but all mining is still done by hand. Pipes and machines exist, so the start of the chain is the missing piece.
PROGRESSION	Iron age onward. The rig takes a drill bit in a tool slot (aluminium, iron or titanium) that sets the tier it can mine, digs a column below its face and pushes drops into its storage or an adjacent pipe.
RECIPE SKETCH	`drill_rig <= basic_machine x2, iron_panels x8, iron_drill_bit x4, microchip x2, energy_cell x4`
BEHAVIOUR	New block (consumer, multi-cell, reserved dig area), server-side block removal through the normal break path (claims, drops), per-entity scheduler, NBT state (progress, cursor, bit wear), UI. Large but central.

D2 Assembler

PRIORITY P2 EFFORT M

IDS	`assembler`
PURPOSE	Every craft is manual. Panels, energy cells and microchips are crafted hundreds of times.
PROGRESSION	Late iron to titanium age. A blueprint is chosen in the UI; ingredients come from pipes or a storage behind it; output goes out the front.
RECIPE SKETCH	`assembler <= crafting_table x1, basic_machine x1, microchip x4, thulium_energy_cell x2`
BEHAVIOUR	New container block that reuses the crafting blueprint data, input/output Storage, pipe endpoint rules and a blueprint picker. Blocks out $-virtual ingredients such as $health.

D3 Sorter and Overflow Pipes

PRIORITY P2 EFFORT S

IDS	`item_pipe_sorter, item_pipe_overflow`
PURPOSE	A pipe has one filter but round-robins its outputs. A pipe network cannot split ores from rock or skip a full machine.
PROGRESSION	Mid game, next to the existing merger.
RECIPE SKETCH	`item_pipe_sorter x1 <= item_pipe x1, microchip x1` `item_pipe_overflow x1 <= item_pipe x1, iron_panels x1`
BEHAVIOUR	Pipe JSON Mode (RoundRobin default, Sorter: filter match goes to Left and the rest Forward, Overflow: Forward first, then sides). Small change in PipeBlock and PipeRegistry.

D4 Incinerator

PRIORITY P3 EFFORT S

IDS	`incinerator`
PURPOSE	Rock and sand overflow is a real problem once automation runs. TODO.txt lists a trash item slot.
PROGRESSION	Mid game. End point of a sorter line.
RECIPE SKETCH	`incinerator <= basic_machine x1, iron_panels x4, compressed_methane x2`
BEHAVIOUR	Consumer block with an IItemTransferEndpoint that deletes inserted items (optional filter), with flame particles while it works.

SECTION 6.5

E. Survival and life support

Reasons to invest in oxygen, healing and mobility beyond the first hour.

E1 Oxygen Tank Upgrades

PRIORITY P2 EFFORT S

IDS	`aluminium_oxygen_tank, titanium_oxygen_tank`
PURPOSE	Equipment raises max health and max power, but never max oxygen. Long EVA trips depend on stacks of compressed oxygen.
PROGRESSION	Two tiers: aluminium (+50) early, titanium (+150) for deep trips to heavy asteroids.
RECIPE SKETCH	`aluminium_oxygen_tank <= aluminium_ingot x4, glass x1, compressed_oxygen x2` `titanium_oxygen_tank <= titanium_ingot x4, thulium_lens x1, compressed_oxygen x4`
BEHAVIOUR	Add an Oxygen bonus (default 0) to equipment, like Health and Power, and a new Tank slot or reuse of the Chest slot. Small code.

E2 Medical Bay

PRIORITY P3 EFFORT M

IDS	`medical_bay (2x1x1)`
PURPOSE	Healing is consumable-only. A base offers no comfort beyond oxygen and gravity.
PROGRESSION	After the DNA reconstructor. Heals 2 HP/s to players standing on it while powered and uses a med gel item from a slot every N HP.
RECIPE SKETCH	`medical_bay <= dna_reconstructor x1, titanium_panels x4, med_kit x4, microchip x2`
BEHAVIOUR	Consumer block with a trigger volume, a slot, sound and particles; healing on the server.

E3 Jetpack Booster

PRIORITY P2 EFFORT M

IDS	`ion_jetpack (new Thrusters slot)`
PURPOSE	Movement speed is fixed for the whole game, while asteroids grow up to 64 radius and sectors are huge.
PROGRESSION	Titanium age: +60 % zero-g acceleration and max speed, extra power use while boosting.
RECIPE SKETCH	`ion_jetpack <= titanium_panels x6, thulium_energy_cell x4, liquid_hydrogen x8, microchip x2`
BEHAVIOUR	New equipment slot and stats read by InertiaController; sync through AstronautNetwork and check MovementGuard limits on the server.

E4 Hydroponics Bay

PRIORITY P3 EFFORT M

IDS	`hydroponics_bay, algae, space_brew`
PURPOSE	Beer exists only as spacer loot. Bases have no living parts.
PROGRESSION	Optional comfort chain: water + light -> algae -> compressed oxygen (slow) or space brew (beer). Good decor for stations.

RECIPE SKETCH	`hydroponics_bay <= glass x6, plastic x4, light_green x1`
	`water x1 -> algae x2 (60 s, needs light)`
	`beer x2 <= algae x2, water x1, glass x1`

BEHAVIOUR	Processing subclass that also checks the light level. Mostly data.

SECTION 6.6

F. Ships

The remaining WP10 items from ships-design.md, plus ship-based mining and heavier hull blocks.

F1 Gyroscope

PRIORITY P1 EFFORT M

IDS	`gyroscope (Type gyroscope)`
PURPOSE	Listed in WP10. Large ships turn poorly because they need balanced thruster pairs on every axis.
PROGRESSION	Ships chapter onward. Pure torque, no linear force; power while turning.
RECIPE SKETCH	`gyroscope <= iron_panels x4, microchip x2, thulium_energy_cell x2, cobalt_lens x2` `JSON: Gyroscope { Torque 400, Power 20 }`
BEHAVIOUR	New block registered through IEntityBlockListener in ShipControlSystem. AutoFlight adds its torque to the rotation classes. Tutorial chapter 9 can mention it as optional.

F2 Magnetic Landing Gear

PRIORITY P2 EFFORT M

IDS	`landing_gear`
PURPOSE	A parked ship drifts with damping and gets bumped by collisions. WP10 lists magnetic landing gear.
PROGRESSION	Ships chapter onward. Locks the ship to the asteroid it touches until it is released by a console key.
RECIPE SKETCH	`landing_gear <= iron_panels x4, cobalt_ingot x4, microchip x1`
BEHAVIOUR	Temporary static joint in ShipBody, a bindable trigger and NBT state. Needs care with collisions and multiplayer.

F3 Docking Connector

PRIORITY P3 EFFORT L

IDS	`docking_connector`
PURPOSE	Two ships cannot join, share power or move cargo. A mothership has no gameplay meaning.
PROGRESSION	Late game. Ships with aligned connectors lock together, merge electrical networks and allow pipe transfer through the connector.
RECIPE SKETCH	`docking_connector <= titanium_panels x6, microchip x4, item_pipe x2, cable_block x4`
BEHAVIOUR	Joint between two ShipBody instances, cross-entity electrical and pipe endpoints, save and sync. Large.

F4 Mining Laser Cannon

PRIORITY P2 EFFORT M

IDS	`cannon_mining_laser`
PURPOSE	Ships cannot mine, so flying is only travel and combat. Ships and mining are the two big systems and they are not linked.
PROGRESSION	Titanium age. A cannon whose projectile damages natural blocks with a drill tier (4) and drops them, so a ship can strip an asteroid surface.
RECIPE SKETCH	`cannon_mining_laser <= titanium_panels x6, iron_lens x4, thulium_energy_cell x4, microchip x2`
BEHAVIOUR	Projectile JSON flag (MiningTier, DropBlocks) handled in the block-hit path of the projectile system; reuses CannonBlock completely. Respects claims and PvP rules.

F5 Composite Hull Blocks

PRIORITY P2 EFFORT XS

IDS	`thulite_block_light/medium/heavy, obscurium_armor_block`
PURPOSE	Titanium heavy (durability 120) is the strongest buildable block, while the anti-frigate cannon is already in the game.
PROGRESSION	Uses the B3 alloys: thulite blocks (durability 150 to 190), obscurium armor (durability 230, PowerToDrill 6). Durability is stored in one byte, so 255 is the limit.
RECIPE SKETCH	`thulite_panels x3 <= thulite_ingot x1, then 1/2/3 panels per block` `obscurium_armor_block <= obscurium_alloy_ingot x2, titanium_panels x2`
BEHAVIOUR	Data-only once B3 exists (until then, use obscurium_ingot directly).

SECTION 6.7

G. End game and the tier-6 gap

Content that gives the obscurium drill a purpose and gives long-term goals to players with a working base.

G1 Tier-6 Ore: Xenite Crystal

PRIORITY P1 EFFORT S

IDS	`xenite_ore, xenite_shard, xenite_crystal`
PURPOSE	The obscurium drill (tier 6) has nothing new to mine: the hardest blocks need power 5. The last drill is a speed upgrade, not a new unlock.
PROGRESSION	A rare glowing ore in the heavy-rock core of heavy and cheese asteroids, or a new crystalline asteroid preset. It gives the end game its own material.

RECIPE SKETCH	`xenite_ore: PowerToDrill 6, LightColor violet, drops xenite_shard x1-2`
	`crusher: xenite_shard x1 -> xenite_dust x2; furnace: xenite_dust x1 -> xenite_crystal x1`
	`xenite_cell x1 <= xenite_crystal x1, thulium_energy_cell x2`

BEHAVIOUR	Data-only: block, items, generation vein, recipes and textures. No existing id changes, so no world upgrade step is needed.

G2 Obscurium Equipment Tier

PRIORITY P2 EFFORT S

IDS	`obscurium armor set, obscurium_flashlight, obscurium_scanner, obscurium_generator`
PURPOSE	Armor, flashlights, scanners and suit generators stop at titanium, so mining obscurium gives no personal upgrade.
PROGRESSION	Fourth tier. Armor about 65 % explosion protection (cap 75 %), generator +45 power, scanner 1600 m with mass and ore hints.
RECIPE SKETCH	`obscurium_chest <= obscurium_alloy_ingot x6, titanium_ingot x6, polymer_fiber x6 (same part ratios as other sets)` `obscurium_generator <= obscurium_ingot x4, xenite_cell x1`
BEHAVIOUR	Data-only (equipment Type, ArmorTexture). Check the protection values with DefaultReactorBalanceTests.

G3 Jump Drive

PRIORITY P3 EFFORT L

IDS	`jump_drive (2x2x3)`
PURPOSE	Ships top out at 20 blocks/s while sectors are 8192 blocks. Far biomes and other players' bases are hard to reach.
PROGRESSION	End game. Charges from the ship network (and batteries), then jumps the ship to a waypoint within N sectors and uses xenite cells.
RECIPE SKETCH	`jump_drive <= obscurium_alloy_ingot x12, xenite_cell x8, microchip x8, thruster_ion x1`
BEHAVIOUR	Large: safe target search (SafeSpawnFinder style), sector loading at the target, teleporting the ship and its passengers, multiplayer authority, cooldown and VFX.

G4 Ore Scanner Mode

PRIORITY P2 EFFORT M

IDS	`Radar.ShowOres (scanner equipment option)`
PURPOSE	The analyzer block reports an asteroid's composition, but nothing helps find a vein while mining. Uranium and obscurium sit in small, deep veins and are found by luck.
PROGRESSION	Iron and titanium scanners get a short-range ore ping; G2 scanner gets a longer range and rare-ore filter.
RECIPE SKETCH	`No new crafting. JSON: Radar { Range 1200, ShowMass true, OreRange 24, Ores [uranium_ore, obscurium_ore] }`
BEHAVIOUR	New optional Radar fields and a periodic background scan of nearby chunks (no per-frame allocations), drawn as outlines or markers.

SECTION 7

Suggested roadmap

Four phases: lowest risk first, each one leaves the game in a releasable state

PHASE 1	PHASE 2	PHASE 3	PHASE 4
Fill the gaps	**Processing and power**	**Automation and ships**	**End game**
Data-only	Small to medium code	Medium to large code	Large code plus content
Patch after 0.1.5	Next minor version	Following minor version	0.2.x

Phase 1: Fill the gaps (Patch after 0.1.5, Data-only)

Closes the crate, iron-lens and tier-6 gaps without code risk. Every item is a JSON entry, a texture and a recipe.

A1 titanium crate and large crates · A2 fast titanium pipes · A3 mining charge · A7 distinct light dyes · A8 gland and nano med kit · A9 loot tables · C3 solar panel (stage 1) · G1 xenite crystal ore

Phase 2: Processing and power (Next minor version, Small to medium code)

Makes a base self-sufficient: water -> oxygen + hydrogen -> power stored in batteries. Rock gets value. Tutorial chapter 6 can gain an optional electrolyzer step.

B1 electrolyzer (+ Byproduct field) · B2 chemical reactor · B6 regolith sifter · C1 battery bank · C2 hydrogen fuel cell · D3 sorter and overflow pipes · A4 regolith bricks · A5 slabs and grating

Phase 3: Automation and ships (Following minor version, Medium to large code)

Closes the automation loop (mine -> sort -> process -> craft -> dispose) and gives ships an economic role.

D1 automated drill rig · D2 assembler · D4 incinerator · F1 gyroscope · F4 mining laser cannon · F2 magnetic landing gear · E1 oxygen tanks · E3 jetpack booster · G4 ore scanner mode

Phase 4: End game (0.2.x, Large code plus content)

Long-term goals for players with a working base: new alloys, the strongest hull, travel across sectors.

B3 alloy smelter and alloys · F5 composite hull blocks · G2 obscurium equipment tier · B4 enrichment centrifuge · C4 power switch and monitor · G3 jump drive · F3 docking connector · E2 medical bay · E4 hydroponics · A6, A10, B5

Priority and effort matrix

	Effort XS	Effort S	Effort M	Effort L
P1	A1, A2, A3	B1, G1	C1, C2, F1	D1
P2	A4, A8, C3, F5	A5, A9, B2, B4, B6, D3, E1, G2	B3, C4, D2, E3, F2, F4, G4	—
P3	A6, A7	A10, D4	B5, E2, E4	F3, G3

Top-left is the best value: P1 ideas that need only JSON. Top-right items (D1 drill rig) are worth their size because they unlock a core pillar.

SECTION 8

Implementation notes

Project rules that every idea above must respect

Topic	Note
Save compatibility	New JSON sections and NBT data need defaults (no Byproduct, no Fuel, no battery_data = current behaviour). New ids need no upgrade step; renaming or removing ids (for example resizing crates in place) needs a step in Game/Versioning/Steps and a test.
Byte limits	Block Durability and Mass are stored in one byte each (max 255), so composite hull blocks stay at or below 230. Max 4096 block types and max stack 255.
Balance tests	DefaultCrusherRecipesTests: crushing never returns more metal than the craft costs. DefaultReactorBalanceTests: explosion damage against armor. DefaultDescriptionsTests: descriptions without numbers.
Tutorials	Ideas that touch tutorial items (compressed oxygen, liquid hydrogen, power, light_white, crafting table, furnace) keep the current hand recipes. New beginner features (battery, electrolyzer, gyroscope) get optional tutorial steps; rewards must never be items a task asks for.
Hot paths	Drill rig, ore scanner and fuel generators run on the server tick or worker pool: no LINQ, no per-frame allocations, reuse pools. GL work stays on the main thread.
Multiplayer	Every new state (battery charge, drill rig progress, switch state, landing-gear lock) is server authoritative and syncs through INetworkStateBlock / BlockStateNetwork, like the existing machines.
Engine boundary	All ideas are game-side (Spacebox). Only a generic feature such as a multi-output processing helper could move into Engine if it has no game references.
Content workflow	Use the add-block and add-item skills; update spec.md (block types table), DONE.txt and the GameSet editor schema (JSON DTO fields appear there automatically through reflection).

Sources read for this document: CLAUDE.md, spec.md, ships-design.md, tutorial-design.md, Spacebox/TODO.txt, Spacebox/DONE.txt, GameSets/Default (blocks.json, items.json, crafting.json, recipes.json, disassembly.json, config.json, ships.json, Generation, Loot, Tutorials) and the block behaviours in Spacebox/Game/Generation/Blocks.